TRIAX

PRICE CHECKER

PRODUCT REQUIREMENTS DOCUMENT

Draft v1.0 · 27 October 2026

Prepared from the recovered project brief and clearly labelled product assumptions

Recovered project direction
A website or application that helps buyers determine a reasonable market price before purchasing, compare available offers, and make more informed buying decisions.

CONFIDENTIAL — INTERNAL PRODUCT PLANNING

Document Control

Field	Details
Product	Price Checker (working title; final product name not confirmed)
Organisation	Triax Technologies Limited (project context recovered from prior discussion)
Document type	Product Requirements Document (PRD) — draft for review
Version / date	v1.0 / 27 October 2026
Product owner / sponsor	To be confirmed by Triax
Technical owner	CTO / engineering lead to confirm
Status	Draft — scope and assumptions require stakeholder validation

Important Note: Recovered Facts vs Proposed Requirements

This document reconstructs a useful working PRD from the available project history. It does not claim that every feature below was part of the original CTO instruction. The recovered brief confirms the core concept, buyer-focused purpose, and intent to research competitors and differentiation. Detailed features, architecture, launch market, and success targets below are proposed for discussion.

Label	Meaning
Recovered	Supported by the available previous conversation: Price Checker was assigned as a website/app concept, intended to help buyers check reasonable prices before buying; competitor research and differentiation were discussed.
Proposed	A recommended product requirement or design choice that has not yet been confirmed by the CTO or product owner.
Decision needed	A material product or technical question that should be answered before implementation.

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1. Executive Summary and Product Vision

Product vision (proposed): Help people make confident purchasing decisions by showing what a product costs across credible sellers and/or market sources, explaining how current and reliable those prices are, and making it easier to spot a potentially unfair offer.

The working concept is a consumer-facing website or application. The initial market and product categories are not confirmed. This draft assumes Nigeria is the first market because the project is being developed in a Nigerian business context; this is a recommendation, not a recovered requirement.

Product principle: Never imply that a displayed price is a definitive market truth when the underlying evidence is limited. Every price insight should communicate source, date, location where relevant, and confidence or coverage.

2. Problem Statement and Opportunity

Buyers often have limited visibility into what a product should reasonably cost. Prices may differ by seller, location, product condition, model, delivery charges, availability, and the date of the quote. Without a simple comparison process, a buyer may overpay, mistake a suspiciously low price for a bargain, or spend too much time checking multiple sources.

2.1 Problem to solve

- Price discovery is fragmented across websites, shops, marketplaces, social channels, and informal recommendations.
- Product listings may use inconsistent names, specifications, pack sizes, models, or conditions, making comparisons unreliable.
- Prices change over time; old data can mislead buyers if it is not visibly dated.
- A lowest-price result alone may hide delivery fees, stock issues, seller risk, or differences in product specifications.
- Buyers need a clear explanation of why a price appears typical, high, or low—not merely a number.

2.2 Opportunity

A trusted price-checking service can reduce search effort, increase price transparency, and create a useful data asset over time. Its defensibility would depend on the quality, freshness, coverage, and provenance of its price data—not just its interface.

3. Goals, Non-Goals and Success Metrics

3.1 Proposed goals

- Allow a user to find a product and view comparable prices from identifiable sources.
- Show the observed price, source, observation time, and relevant product/location details.
- Explain a price range or price assessment only when there is sufficient comparable data.
- Help users compare total cost where delivery or other mandatory charges are available.
- Make reporting an incorrect price or listing straightforward.
- Build an architecture that can add categories, sources, and geographic coverage incrementally.

3.2 Non-goals for the initial release

- Do not build a full e-commerce checkout, inventory system, or logistics network.

- Do not guarantee that a seller is legitimate or that a product is authentic unless a separate verification process supports that claim.
- Do not promise real-time prices unless a source actually provides and permits real-time access.
- Do not make unsupported claims that a price is fraudulent, exploitative, or the single correct market price.
- Do not add AI price predictions or automated buying without validating data quality and user demand.

3.3 Proposed initial success metrics

Metric	Proposed definition	Initial target (to validate)
Search success rate	Share of searches that lead to a product result or useful no-result guidance.	≥ 70% after initial category coverage is established
Price coverage	Share of supported product pages with at least two comparable, recent observations.	Set baseline per launch category; aim to improve monthly
Price freshness	Share of visible observations within category-specific freshness windows.	≥ 80% within defined freshness window
Comparison engagement	Share of product viewers who open or interact with price comparisons.	Establish baseline during beta
Correction resolution	Share of validated price reports resolved within the published service target.	≥ 90% within target SLA
Trust / usefulness	User-reported helpfulness of the price information.	Establish baseline in beta

Targets above are suggested starting points, not commitments. Final targets should be agreed after category selection, source feasibility testing, and a small beta.

4. Target Users and Key Use Cases

4.1 Primary user groups (proposed)

User	Need	Example
Everyday buyer	Check whether an asking price is reasonable.	Before buying a phone, appliance, or household item.
Budget-conscious shopper	Compare total cost and find a better offer.	Compare seller price plus delivery charges.
Small business buyer	Benchmark purchase costs and compare suppliers.	Check prices for equipment or recurring supplies.
Seller / merchant	Submit or correct product and price information.	Provide an updated listing or offer.
Operations / data reviewer	Maintain trustworthy data and resolve reports.	Review stale, duplicated, or suspicious listings.

4.2 Core use cases

- Search by product name, brand, model, category, or a supported identifier such as SKU/GTIN where available.
- Filter and compare results by location, seller, condition, availability, and total price where data supports those fields.
- Open a product page to inspect price observations and their source timestamps.
- See a clearly qualified indication such as “within observed range” or “above observed range” only when enough comparable data exists.
- Report a price that is outdated, inaccurate, duplicated, or attached to the wrong product.
- For a later release, submit merchant offers through a verified seller workflow.

5. Scope and Release Strategy

Because the original scope is not fully recovered, this PRD recommends a narrow MVP rather than attempting every market and category immediately.

5.1 MVP (proposed)

- Responsive web experience, mobile-first.
- A small number of validated product categories selected with the CTO/product owner.
- Search and product detail pages.
- A price comparison list with source, observed date, currency, product match attributes, and available seller/location details.
- Basic filters and sorting where data quality allows.
- A price-range summary only where minimum evidence and comparability rules are met.
- Admin interface or controlled internal workflow for reviewing sources, imports, corrections, and reports.
- Basic analytics, logging, error handling, and privacy information.

5.2 Later releases (subject to validation)

- User accounts, saved products, and price alerts.
- Merchant accounts, self-service price submissions, and verification workflows.
- Price history charts and category-level trends.
- Location-aware physical-market observations and field-research submissions.
- Image-based product identification or barcode scanning.
- Additional countries, languages, and currencies.
- Personalised recommendations or predictive features after data quality is proven.

5.3 Explicit scope decision

The largest unresolved question is whether Price Checker is primarily an online-store comparison service, a physical/local-market price guide, or a hybrid. This affects data acquisition, product matching, freshness expectations, and operational costs. The MVP should not assume a hybrid model until the product owner confirms it.

6. Functional Requirements

ID / Priority	Requirement	Acceptance criteria
FR-01 / Must	Product search	Users can search supported products by name and available identifiers. Empty, loading, error, and no-result states are handled.
FR-02 / Must	Product matching	Results distinguish model, variant, size/capacity, condition, and pack quantity where applicable; ambiguous matches are not silently merged.
FR-03 / Must	Price comparison	A product page lists comparable observations with amount, currency, source/seller, observed time, and known availability/location fields.
FR-04 / Must	Source transparency	Every displayed observation links to or identifies its source where permitted, and shows when the price was last observed.
FR-05 / Must	Sorting	Users can sort by price and, when supported, recency or relevance. Total-cost sorting is used only when comparable fees are available.
FR-06 / Must	Price assessment	A range or assessment is shown only when enough recent, comparable observations exist; otherwise the page explains that evidence is insufficient.
FR-07 / Must	Out-of-date data	Stale observations are visibly marked or excluded from current-price calculations according to category rules.
FR-08 / Must	Report an issue	Users can report incorrect price, wrong product, unavailable item, duplicate, or broken source. Submission receives a confirmation.
FR-09 / Must	Admin review	Authorised staff can inspect observations, sources, reports, and ingestion errors; changes are auditable.
FR-10 / Must	Responsive accessibility	Core journeys work on common mobile and desktop widths and can be navigated using a keyboard.
FR-11 / Should	Category and location filters	Users can narrow results where category/location data exists; unavailable filters are not shown as if they work.
FR-12 / Should	Price history	A history view is available only for products with sufficient time-series observations.
FR-13 / Later	Saved items and alerts	Users can save products and opt in to alerts with clear frequency and notification controls.
FR-14 / Later	Merchant submissions	Merchants can submit or update offers after identity and data-quality checks are defined.

7. Price Data Quality and Trust

7.1 Minimum price-observation fields

Field	Purpose
Product identity	Canonical product and variant; brand/model; condition; size/capacity or pack quantity where applicable.
Observed price	Amount and currency; whether the value is a list price, sale price, or quote.
Source	Seller/source name and source URL or internal reference, subject to permitted use.
Observation timestamp	When the price was observed; ingestion timestamp stored separately.
Geography	Country and, where relevant, state/city/market or delivery destination.
Availability	In stock, out of stock, unknown, or source-provided status.
Additional cost	Delivery, service, or mandatory fees if known; do not infer missing costs.
Evidence / provenance	How the data was obtained, source reliability, and any validation status.
Status	Active, stale, rejected, removed, or under review, with reason and audit trail.

7.2 Price-range calculation (proposed)

- Compare like with like: same model/variant, condition, quantity, currency, and relevant location.
- Exclude observations that are stale, invalid, duplicated, suspicious, or not comparable.
- Use robust statistics such as median and percentile range rather than relying solely on the minimum or average.
- Set minimum evidence thresholds by category; if thresholds are not met, display “Not enough recent comparable prices to estimate a reliable range.”
- Explain that the range represents observed data, not a guaranteed fair price or official valuation.
- Do not compare prices across currencies without recording the exchange-rate source and conversion time.

7.3 Suggested labels

Label	Rule of use
Within observed range	The price falls inside the calculated range from sufficient comparable observations.
Above observed range	The price exceeds the selected upper threshold; show sample size, range, and date context.
Below observed range	The price is below the selected lower threshold; do not automatically call it a bargain or scam.
Insufficient data	Evidence is too sparse, stale, or inconsistent to support a responsible assessment.

The thresholds and minimum sample sizes must be determined through product testing and category-specific analysis. Do not hard-code a universal “fair price” rule for every product.

8. User Journeys and Acceptance Criteria

8.1 Journey A — Find a product

- User opens the homepage and enters a product query.
- System returns relevant products with distinguishing attributes.
- User selects the correct model/variant.
- System displays available price observations and freshness information.

Acceptance: A user can complete the journey on a mobile browser; search states are clear; product variants are distinguishable; no price is displayed without source and observation context.

8.2 Journey B — Assess a quoted price

- User opens a product page or enters a quoted price in a later enhancement.
- System compares the quote against recent, comparable observations.
- System presents the observed range and sample context, or explains why it cannot produce a reliable assessment.
- User can open source details and report a mismatch.

Acceptance: The assessment is reproducible from recorded observations; insufficient data does not generate a confident label; stale or mismatched prices do not silently influence the result.

8.3 Journey C — Report incorrect information

- User selects a report reason and optionally provides details.
- System validates the submission and records a case.
- Operations reviews and resolves the case, retaining an audit trail.
- Where contact information is collected, the user is told how it will be used.

Acceptance: Duplicate or abusive submissions are rate-limited; staff can see case status; the user receives a clear submission confirmation.

9. Non-Functional Requirements

Area	Proposed requirement
Performance	Target a fast mobile experience on variable network quality; define and measure p75 page-load and API latency during beta.
Availability	Agree a service target before public launch; communicate planned maintenance and degrade gracefully if data sources fail.
Scalability	Separate search/read traffic from ingestion and admin workloads; use pagination and caching where appropriate.
Accessibility	Target WCAG 2.2 AA for core journeys; semantic headings, keyboard support, visible focus, labels, and adequate contrast.
Security	Use server-side validation, least-privilege access, secure session handling, rate limits, dependency updates, and protected secrets.
Data integrity	Use stable product IDs, validation constraints, deduplication, timestamps, and auditable changes.
Maintainability	Automated tests, documented setup, migrations, environment configuration, monitoring, and backup/restore procedures.
Localization	Use explicit currency codes and locale-aware formatting; avoid assuming that all prices are in NGN if expansion is planned.
Resilience	Search and product pages should explain missing or delayed data rather than presenting ingestion failures as valid prices.

10. Data Model and Integrations

10.1 Suggested core entities

Entity	Key fields (illustrative)
Product	id, canonical_name, brand, category_id, identifiers, created_at, status
ProductVariant	id, product_id, model, attributes/specifications, condition, pack_size
Seller / Source	id, name, source_type, source_url, region, terms_review_status, reliability metadata
PriceObservation	id, variant_id, source_id, amount, currency, observed_at, ingested_at, availability, location, fees, status
PriceAssessment	id, variant_id, calculated_range, sample_size, method_version, calculated_at, freshness status
UserReport	id, observation_id, reason, details, status, submitted_at, resolution metadata
AdminAuditEvent	id, actor_id, action, entity_type, entity_id, timestamp, change summary

10.2 Potential data sources

- Official seller or retailer APIs where available and permitted.
- Merchant-provided catalog/price feeds under agreed terms.
- Publicly accessible pages only where collection and reuse comply with applicable law, site terms, and technical restrictions.
- Structured manual price checks or field research if the product includes physical markets.
- User-submitted observations only with moderation, provenance, anti-abuse checks, and visible confidence controls.

Do not assume that a website can be scraped legally, reliably, or economically. Each source needs a feasibility and permissions review before it is committed to the MVP.

11. Suggested Technical Architecture

The following is a technology-neutral reference architecture. It is not a recovered or mandated stack.

Layer	Responsibility
Web client	Responsive search, product details, comparisons, reports, and accessible states.
Application API	Search, product resolution, comparison, reports, admin workflows, authentication if required.
Product catalogue	Canonical products and variants with controlled matching rules.
Ingestion workers	Scheduled or event-driven source imports, validation, normalization, deduplication, and error handling.
Relational database	Products, observations, sources, reports, users/admins, and audit events.
Search index (optional)	Fast text search and typo tolerance if database search is insufficient.

Layer	Responsibility
Cache / queue (optional)	Protect source systems, schedule ingestion, and improve performance at scale.
Observability	Application logs, ingestion metrics, error alerts, uptime checks, and audit review.

Architecture decision should follow a short technical discovery: expected traffic, initial categories, data sources, budget, deployment environment, and team familiarity. Avoid adding a search engine, queue, or microservices unless the measured need justifies the operational cost.

12. Privacy, Security and Abuse Prevention

- Collect the minimum personal data needed. Guest browsing should remain possible unless a feature genuinely requires an account.
- Publish a privacy notice explaining collected data, purpose, retention, sharing, user rights, and contact route.
- Protect admin routes with strong authentication, role-based authorization, and audit logging.
- Validate and sanitize user-submitted text and URLs; prevent injection, cross-site scripting, and server-side request forgery risks.
- Rate-limit searches, reports, sign-ups, and submission endpoints; use anti-automation controls where appropriate.
- Do not expose private source credentials, API keys, internal review notes, or unpublished merchant data.
- Define a process for copyright, takedown, merchant disputes, inaccurate pricing, and data-removal requests.
- Review relevant Nigerian data-protection requirements and any other jurisdiction-specific rules before launch.

13. Analytics and Operational Metrics

Metric	Why it matters
Search volume and zero-result queries	Shows demand, catalogue gaps, and wording users actually use.
Search-to-product click-through	Measures relevance of search results.
Price observations per product	Measures comparison depth and coverage.
Observation freshness by source/category	Detects stale data and ingestion problems.
Product-match correction rate	Reveals faulty variant matching.
Report rate and resolution time	Measures data quality and operational responsiveness.
Source failure rate	Highlights broken imports, blocked access, or upstream changes.
Latency and error rate	Measures reliability of the core experience.
Repeat use and saved-product use	Signals whether the service provides ongoing value, if accounts are later introduced.

Analytics should avoid unnecessary personal tracking. Establish retention periods and access controls for event data.

14. Risks, Dependencies and Open Questions

14.1 Key risks

Risk	Impact	Mitigation
Insufficient price coverage	The product cannot produce useful comparisons.	Start with a narrow category/geography; validate data supply before broad UI development.
Stale or inaccurate prices	Users lose trust or make poor decisions.	Show observation dates, source status, freshness rules, and correction workflow.
Incorrect product matching	Unfair or meaningless comparisons.	Use variant attributes, confidence checks, and manual review for ambiguous matches.
Source access restrictions	Imports may stop or be prohibited.	Review terms and permissions; prefer APIs/feeds and diversify sources.
False “deal” or scam signals	Reputational and potential legal risk.	Use qualified language, explain evidence, and avoid unsupported accusations.
Scope too broad	Long delivery time and weak initial coverage.	Select one category and market for the MVP; gate expansion on metrics.
Operational workload	Manual review becomes expensive.	Measure review volume and automate only after reliable rules are established.

14.2 Decisions needed from the product owner / CTO

- Is the product online-only, physical-market-focused, or hybrid?
- Which product category or categories should launch first?
- Which city, region, or country is the initial target?
- Who supplies the data: retailers, APIs, manual checks, user submissions, or a combination?
- Will merchants be able to submit listings in the MVP?
- Is the first release a public website, mobile app, or responsive web app?
- What is the approved budget, delivery window, and hosting environment?
- What does management consider the product's main differentiator?
- Who owns source permissions, privacy/legal review, and dispute resolution?

15. Delivery Roadmap and Definition of Done

Phase	Work	Exit criteria
Phase 0 — Discovery	Confirm scope, interview target users, choose initial category/geography, test source access and data quality.	Approved scope and evidence that useful price observations can be obtained lawfully and consistently.
Phase 1 — Foundation	Define product schema, source model, search strategy, ingestion prototype, and basic admin controls.	A repeatable pipeline imports and validates sample observations with provenance.
Phase 2 — MVP build	Implement search, product details, comparisons, freshness labels, reporting, and responsive UX.	Critical user journeys pass acceptance tests on mobile and desktop.
Phase 3 — Closed beta	Test with real users and a limited catalogue; measure search success, coverage, freshness, and errors.	Known data-quality issues documented; key thresholds and launch gates agreed.
Phase 4 — Public launch	Deploy monitoring, privacy notice, support process, backup/restore, and operational runbook.	Launch checklist approved; no unresolved critical security or data-integrity defects.
Phase 5 — Expansion	Add categories, alerts, merchant tools, history, or local-market collection based on evidence.	Expansion is justified by user demand, data supply, and operational capacity.

15.1 Definition of done for MVP

- Product owner approves the initial scope and supported categories.
- Data-source permissions and acquisition methods are documented and reviewed.
- Every displayed price includes the required source and time context.
- Product matching and price-range rules are tested, including insufficient-data cases.
- Core flows pass functional, responsive, accessibility, and security checks.
- Admin review and audit trail work for price corrections and user reports.
- Monitoring, backup/restore, error handling, and deployment documentation are in place.
- Privacy notice, support contact, and data correction/takedown workflow are ready.
- Beta findings and known limitations are visible to the team and product owner.

Appendix A — Suggested MVP User Stories

ID	User story	Priority
US-01	As a buyer, I want to search for a product so I can find relevant price information.	Must
US-02	As a buyer, I want to confirm the exact model or variant so I do not compare different products.	Must
US-03	As a buyer, I want to see price, source, and observation date so I can judge how useful the information is.	Must
US-04	As a buyer, I want a qualified price range when evidence is sufficient so I can understand the market context.	Must
US-05	As a buyer, I want to know when there is not enough reliable data so I am not misled.	Must
US-06	As a buyer, I want to report incorrect information so the catalogue can improve.	Must
US-07	As an operator, I want to review source failures and user reports so I can maintain data quality.	Must
US-08	As a returning user, I want to save a product and receive an alert when its price changes.	Later

Appendix B — Release-Gate Checklist

- Scope and launch category signed off.
- Data-source feasibility and permissions reviewed.
- Price observation schema and freshness policy approved.
- Minimum evidence thresholds and comparison rules tested.
- No unsupported “fair price,” “cheapest,” or fraud claims.
- Privacy, security, and takedown processes reviewed.
- Mobile performance and accessibility checked.
- Support ownership, monitoring, and incident response assigned.
- Beta success metrics and expansion criteria agreed.

Closing Recommendation

Treat Price Checker first as a data-trust product and second as a comparison interface. A polished interface cannot compensate for stale, mismatched, or thin price data. Validate the initial market and data supply before committing to a broad build, then ship a narrow, dependable MVP and expand only when coverage and user value are demonstrated.

Next action: Review this draft with the CTO/product owner, resolve the open questions in Section 14.2, and issue PRD v1.1 with approved scope, data-source strategy, and measurable launch targets.